

Supply Chain and Warehouse Logistics Network

Phase 1 Conceptual Design

This report presents the conceptual design of a database that tracks inventory across warehouses in different geographical locations, supplier orders, shipments, and delivery events. The model addresses two central requirements: multi-level product structures and restock alerts when inventory falls below its reorder level.

The Phase 1 result defines the system scope, 11 functional requirements, 15 business rules, 13 entities, and 17 relationships with explicit cardinality and participation. It also provides a complete EER model and conceptual logic for recursive product components and inventory alerts.

Information	Details
Course	Database Systems
Topic	6 Supply Chain and Warehouse Logistics Network
Stage	Phase 1 Conceptual Design
Submission	Progress report and EER model

Completed Work

- Defined the problem, users, scope, and assumptions of the system.
- Modelled warehouses, products, suppliers, supplier orders, shipments, tracking events, and inventory.
- Represented multi-level product structures through the recursive ProductComponent entity.
- Specified when a restock alert must be created, retained, closed, and opened again.
- Checked completeness with acceptance cases at the end of the report.

I. Problem Context and Scope

1.1 Problem Statement

The organization operates warehouses in different cities or countries. A product may be stored in several warehouses, a supplier may provide several products, and one supplier order may be delivered through several shipments. Fragmented records make it difficult to confirm the stock held at each location, compare ordered and delivered quantities, track shipments, and identify products that require replenishment.

Some finished products are assembled from components that may contain their own subcomponents. A flat product list cannot represent this structure or support material

requirement calculations. The database must support an unlimited number of component levels while preventing a product from containing itself directly or indirectly.

1.2 Objectives

- Manage warehouses, products, suppliers, and supplier product details in one system.
- Maintain one inventory balance for every warehouse and product combination.
- Track supplier orders, order items, partial shipments, and shipment events over time.
- Represent multi-level product structures with the quantity required at each level.
- Record every stock movement and issue an alert when inventory is below its reorder level.
- Provide a clear EER model that can be mapped to a relational design in Phase 2.

1.3 Scope and Assumptions

Category	Details
In scope	Geographical warehouses; products and suppliers; current supplier prices; supplier orders; partial shipments; tracking events; inventory; stock movements; multi-level product structures; restock alerts.
Outside Phase 1	Customer sales; route optimization; production batches; serial numbers; returns; warehouse transfers; unit conversion; BOM version history; payments and accounting.
Order assumptions	Each order has one supplier and one destination warehouse. A product appears at most once in the same order.
Data assumptions	Each product uses one base unit; prices use one common currency; event times use the same standard time zone.

II. Entities, Attributes, Relationships and Definitions

This section focuses on the definitions required for the Phase 1 conceptual model. Each entity is explained in normal prose, while the attribute, relationship, and function details are kept in tables for easier reading and checking.

2.1 System Definition

The Supply Chain and Warehouse Logistics Network Database manages products across multiple warehouses, suppliers, supplier orders, shipments, inventory transactions, multi-level product structures, and restock alerts. It provides a consistent view of what products exist, where they are stored, which suppliers offer them, what has been ordered, how orders are delivered, how inventory changes, and when stock falls below its reorder level.

2.2 User / Actor Definitions

2.2.1 Purchasing Staff: Employees responsible for **purchasing goods and working with suppliers.**

2.2.2 Warehouse Staff: Employees responsible for **receiving, issuing, and managing goods in the warehouse.**

2.2.3 Logistics Staff: Employees responsible for **shipping, delivery, and tracking goods.**

2.2.4 Inventory Manager: The person responsible for **monitoring inventory levels and deciding when goods need to be replenished.**

2.3 Users and Main Workflow

User Role	Main Responsibility
Purchasing Staff	Manage supplier information, supplier-product offerings and prices, create and manage supplier orders, and monitor order status.
Warehouse Staff	View inventory, confirm shipment receipts, and perform stock issues and adjustments through the stock-movement process.
Logistics Staff	Create shipments, allocate order quantities to shipments, dispatch shipments, confirm arrival, and record tracking events.
Inventory Manager	Review inventory levels, reorder conditions, low-stock alerts, and make replenishment decisions.

2.4 Entity Definitions

2.4.1 Warehouse

Definition: Represents a physical storage location. It stores warehouse identification and location/capacity information.

Entity type: Strong (independent) entity.

Identifier: WarehouseID

Relationship meaning: A Warehouse may have 0..N Inventory records and may be the destination of 0..N SupplierOrders.

2.4.2 Product

Definition: Represents a stocked, purchased, or assembled item.

Entity type: Strong (independent) entity.

Identifier: ProductID

Relationship meaning: A Product may have 0..N Inventory and SupplierProduct records, may appear in many SupplierOrderItems, and may participate as parent or component in ProductComponent.

2.4.3 Inventory

Definition: Represents the current stock balance of one Product at one Warehouse.

Entity type: Associative entity depending on Warehouse and Product.

Identifier: InventoryID

Relationship meaning: Each record belongs to exactly one Warehouse and one Product; the WarehouseID + ProductID pair is unique.

2.4.4 Supplier

Definition: Represents a supplier/product source.

Entity type: Strong (independent) entity.

Identifier: SupplierID

Relationship meaning: A Supplier may offer many Products through SupplierProduct and may be referenced by many SupplierOrders.

2.4.5 SupplierProduct

Definition: Represents a supplier's offering of a product, including current price and lead time.

Entity type: Associative entity (Supplier $\times$ Product).

Identifier: SupplierID + ProductID

Relationship meaning: Resolves the many-to-many Supplier–Product relationship.

2.4.6 SupplierOrder

Definition: Represents a purchase order placed with a supplier.

Entity type: Strong entity (transaction header).

Identifier: OrderID

Relationship meaning: Each order has exactly one Supplier and one destination

Warehouse; a submitted order must contain at least one item.

2.4.7 SupplierOrderItem

Definition: Represents one product line within a supplier order.

Entity type: Weak entity depending on SupplierOrder.

Identifier: OrderItemID

Relationship meaning: Each item belongs to exactly one order and one Product; a Product cannot repeat in the same order; UnitPrice is the historical agreed price.

2.4.8 Shipment

Definition: Represents a physical delivery made against a supplier order.

Entity type: Strong entity (transaction header).

Identifier: ShipmentID

Relationship meaning: A SupplierOrder may have 0..N Shipments; each Shipment belongs to one order and inherits its destination Warehouse.

2.4.9 ShipmentItem

Definition: Represents the quantity of an order line allocated to a shipment.

Entity type: Weak/associative entity depending on Shipment and SupplierOrderItem.

Identifier: ShipmentItemID

Relationship meaning: Supports partial delivery; total non-cancelled allocations cannot exceed the ordered quantity.

2.4.10 TrackingEvent

Definition: Represents a shipment tracking/status event.

Entity type: Weak entity depending on Shipment.

Identifier: TrackingEventID

Relationship meaning: Each event belongs to one Shipment; a Shipment may have 0..N events.

2.4.11 ProductComponent

Definition: Represents a direct parent–component relationship in a multi-level product structure.

Entity type: Recursive associative entity (Product $\times$ Product).

Identifier: ParentProductID + ComponentProductID

Relationship meaning: The same Product entity appears in parent and component roles; direct or indirect cycles are not allowed.

2.4.12 StockMovement

Definition: Represents a successful inventory transaction and preserves stock-change history.

Entity type: Weak entity depending on Inventory; optionally ShipmentItem.

Identifier: MovementID

Relationship meaning: Receipt movements may reference a ShipmentItem; other movement types do not.

2.4.13 RestockAlert

Definition: Represents one low-stock alert period.

Entity type: Weak entity depending on Inventory.

Identifier: AlertID

Relationship meaning: An Inventory record may have at most one unresolved alert; a later low-stock period creates a new alert after the previous one is resolved.

2.5 Attribute Definitions and Data Dictionary

2.5.1 Warehouse

Attribute	Data Type	Key	Null	Description / Constraints
WarehouseID	INT	PK	NOT NULL	Unique identifier of a warehouse.
WarehouseName	VARCHAR(100)	—	NOT NULL	Name of the warehouse.
Address	VARCHAR(200)	—	NOT NULL	Physical address of the warehouse.
City	VARCHAR(100)	—	NOT NULL	City where the warehouse is located.
Country	VARCHAR(100)	—	NOT NULL	Country where the warehouse is located.
CapacityM3	DECIMAL(12,2)	—	NOT NULL	Maximum storage capacity in cubic meters. Must not be negative.

2.5.2 Product

Attribute	Data Type	Key	Null	Description / Constraints
ProductID	INT	PK	NOT NULL	Unique identifier of a product.
ProductName	VARCHAR(150)	—	NOT NULL	Name used to identify the product.
Description	TEXT	—	NULL	Additional descriptive information about the product.
Unit	VARCHAR(20)	—	NOT NULL	Base unit used to measure the product quantity.
ProductType	VARCHAR(50)	—	NULL	Type or category of the product. Controlled list is finalized in Phase 2.

2.5.3 Inventory

Attribute	Data Type	Key	Null	Description / Constraints
InventoryID	INT	PK	NOT NULL	Unique identifier of an inventory record.
WarehouseID	INT	FK	NOT NULL	References Warehouse. Each inventory belongs to exactly one warehouse.
ProductID	INT	FK	NOT NULL	References Product. Each inventory belongs to exactly one product.
Quantity	DECIMAL(12,2)	—	NOT	Current quantity available. Cannot be

			NULL	negative.
ReorderLevel	DECIMAL(12,2)	—	NOT NULL	Minimum inventory threshold. Cannot be negative.
LastUpdated	DATETIME	—	NOT NULL	Date and time when the inventory balance was most recently changed.

2.5.4 Supplier

Attribute	Data Type	Key	Null	Description / Constraints
SupplierID	INT	PK	NOT NULL	Unique identifier of a supplier.
SupplierName	VARCHAR(150)	—	NOT NULL	Name of the supplier.
ContactName	VARCHAR(100)	—	NULL	Name of the main contact person.
Phone	VARCHAR(30)	—	NULL	Supplier's contact telephone number.
Email	VARCHAR(150)	—	NULL	Supplier's contact email address.
Address	VARCHAR(200)	—	NULL	Supplier's business address.

2.5.5 SupplierProduct

Attribute	Data Type	Key	Null	Description / Constraints
SupplierID	INT	PK, FK	NOT NULL	References Supplier and forms part of the composite primary key.
ProductID	INT	PK, FK	NOT NULL	References Product and forms part of the composite primary key.
UnitPrice	DECIMAL(12,2)	—	NOT NULL	Current price offered by the supplier.
LeadTimeDays	INT	—	NOT NULL	Expected number of days required for delivery.

2.5.6 SupplierOrder

Attribute	Data Type	Key	Null	Description / Constraints
OrderID	INT	PK	NOT NULL	Unique identifier of a supplier order.
OrderDate	DATE	—	NOT NULL	Date on which the supplier order is created.
ExpectedDeliveryDate	DATE	—	NULL	Expected delivery date at the destination warehouse.
Status	VARCHAR(20)	—	NOT NULL	Current lifecycle state of the supplier order.
SupplierID	INT	FK	NOT NULL	References Supplier. Each order refers to exactly one supplier.
WarehouseID	INT	FK	NOT NULL	References Warehouse. Each order has exactly one destination warehouse.

2.5.7 SupplierOrderItem

Attribute	Data Type	Key	Null	Description / Constraints
OrderItemID	INT	PK	NOT NULL	Unique identifier of an individual order line.
OrderID	INT	FK	NOT NULL	References SupplierOrder.

ProductID	INT	FK	NOT NULL	References Product.
Quantity	DECIMAL(12,2)	—	NOT NULL	Quantity of the product requested in the order line.
UnitPrice	DECIMAL(12,2)	—	NOT NULL	Historical price agreed when the order was created.

2.5.8 Shipment

Attribute	Data Type	Key	Null	Description / Constraints
ShipmentID	INT	PK	NOT NULL	Unique identifier of a shipment.
OrderID	INT	FK	NOT NULL	References SupplierOrder. One order may have multiple shipments.
TrackingNumber	VARCHAR(50)	—	NULL	External tracking reference assigned to the shipment.
ShipmentDate	DATE	—	NULL	Date when the shipment is dispatched.
EstimatedArrival	DATETIME	—	NULL	Current expected arrival date and time.
ActualArrival	DATETIME	—	NULL	Confirmed arrival date and time. Empty until arrival is confirmed.
Status	VARCHAR(20)	—	NOT NULL	Current logistics state of the shipment.

2.5.9 ShipmentItem

Attribute	Data Type	Key	Null	Description / Constraints
ShipmentItemID	INT	PK	NOT NULL	Unique identifier of a shipment line.
ShipmentID	INT	FK	NOT NULL	References Shipment.
OrderItemID	INT	FK	NOT NULL	References SupplierOrderItem. Must belong to the same supplier order as the shipment.
Quantity	DECIMAL(12,2)	—	NOT NULL	Quantity allocated to this shipment. Must be positive. Total allocation cannot exceed the originally ordered quantity.

2.5.10 TrackingEvent

Attribute	Data Type	Key	Null	Description / Constraints
TrackingEventID	INT	PK	NOT NULL	Unique identifier of a tracking event.
ShipmentID	INT	FK	NOT NULL	References Shipment.
EventTime	DATETIME	—	NOT NULL	Date and time when the tracking event occurs.
Location	VARCHAR(150)	—	NULL	Geographical location where the event is recorded.
Status	VARCHAR(30)	—	NOT NULL	Shipment status reported by the tracking event.

Description	TEXT	—	NULL	Additional information about the event or handling activity.
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2.5.11 ProductComponent

Attribute	Data Type	Key	Null	Description / Constraints
ParentProductID	INT	PK, FK	NOT NULL	References Product as the parent/assembled product.
ComponentProductID	INT	PK, FK	NOT NULL	References Product as the component product.
QuantityRequired	DECIMAL(12,2)	—	NOT NULL	Number of component units required to produce one parent product. Must be greater than zero.

2.5.12 StockMovement

Attribute	Data Type	Key	Null	Description / Constraints
MovementID	INT	PK	NOT NULL	Unique identifier of an inventory transaction.
InventoryID	INT	FK	NOT NULL	References Inventory.
ShipmentItemID	INT	FK	NULL	Optional reference to ShipmentItem. Used only for Receipt movements.
MovementTime	DATETIME	—	NOT NULL	Date and time when the inventory transaction is recorded.
MovementType	VARCHAR(20)	—	NOT NULL	Category of inventory change, e.g. Opening, Receipt, Issue, Adjustment.
QuantityDelta	DECIMAL(12,2)	—	NOT NULL	Signed inventory change. Positive increases inventory; negative decreases inventory.

2.5.13 RestockAlert

Attribute	Data Type	Key	Null	Description / Constraints
AlertID	INT	PK	NOT NULL	Unique identifier of a low-stock alert period.
InventoryID	INT	FK	NOT NULL	References Inventory.
AlertDate	DATETIME	—	NOT NULL	Date and time when the low-stock condition is first detected.
QuantityAtAlert	DECIMAL(12,2)	—	NOT NULL	Inventory quantity captured when the alert is created. Historical snapshot.
ReorderLevelAtAlert	DECIMAL(12,2)	—	NOT NULL	Reorder level captured when the alert is created. Historical snapshot.
ResolvedAt	DATETIME	—	NULL	Date and time when the alert is resolved. NULL means unresolved.
Message	VARCHAR(200)	—	NULL	Human-readable explanation of the low-stock condition.

2.6 Function Definitions

ID	Function	Actor / Executor	Definition / Purpose	Related FR / BR
F-WH1	Register / update warehouse	Purchasing Staff	Create/edit warehouse identification and location/capacity information.	FR01
F-WH2	View warehouse inventory	Warehouse Staff; Inventory Manager	View current Quantity and ReorderLevel for products at a warehouse.	FR07
F-WH3	Confirm shipment receipt	Warehouse Staff	Confirm receipt of a ShipmentItem; creates Receipt StockMovement and updates Inventory.	FR08
F-WH4	Act as order destination	System	Set one destination Warehouse for an order; shipments inherit it.	FR04, BR06
F-PR1	Register / update product	Purchasing Staff	Manage product name, description, base unit and type.	FR02
F-PR2	Define product structure (BOM)	Authorized product-management role	Add/remove parent-component links with required quantity; system checks cycles.	FR09, FR10
F-PR3	View stock across warehouses	Inventory Manager	Check a product's Quantity across warehouses.	FR07
F-IN1	Maintain inventory balance	System	Maintain current Quantity and ReorderLevel for each warehouse-product pair.	FR07
F-IN2	Record stock movement	System	Create one StockMovement and update Quantity for each successful stock operation.	FR08
F-IN3	Evaluate restock condition	System	After Quantity/ReorderLevel changes, open or close a RestockAlert as required.	FR11
F-SU1	Register / update supplier	Purchasing Staff	Manage supplier contact information.	FR03
F-SU2	Manage product catalog & pricing	Purchasing Staff	Manage supplier-product offerings, current UnitPrice and LeadTimeDays.	FR03, BR02
F-SU3	Receive supplier orders	System / business process	Ensure an order has one supplier and that the supplier offers every ordered product.	FR04, BR05
F-SP1	Quote / update price	Purchasing Staff	Record the supplier's current UnitPrice for a product.	FR03
F-SP2	Record lead time	Purchasing Staff	Record expected LeadTimeDays for delivery planning.	FR03
F-SO1	Create order	Purchasing Staff	Open a draft order for one supplier and one destination warehouse.	FR04
F-SO2	Add / remove order items	Purchasing Staff	Add product lines and quantities; prevent duplicate product lines.	FR04, BR04

F-SO3	Submit order	Purchasing Staff	Change Draft to Submitted only when at least one item exists.	BR04
F-SO4	Track order status	Purchasing Staff; Inventory Manager	Monitor order status as shipments are created and received.	FR04
F-OI1	Record order line	System	Store product, ordered quantity and agreed historical price.	FR04, BR05
F-OI2	Check allocation limit	System	Ensure shipment allocations never exceed ordered quantity.	BR08
F-SH1	Create shipment	Logistics Staff	Create a shipment against a supplier order.	FR05
F-SH2	Allocate order item to shipment	Logistics Staff	Allocate order quantities across shipments within the ordered limit.	FR05, BR08
F-SH3	Dispatch shipment	Logistics Staff	Dispatch a shipment only after it has at least one ShipmentItem.	BR07
F-SH4	Confirm arrival	Warehouse Staff / Logistics Staff	Record ActualArrival and update shipment status when delivered.	FR06
F-SI1	Validate matching order	System	Reject a ShipmentItem linked to an order item from another order.	BR07
F-SI2	Trigger receipt	Warehouse Staff	Receive a ShipmentItem at most once and create one Receipt StockMovement.	FR08, BR13
F-TE1	Log tracking event	Logistics Staff	Append chronological status/location information to a shipment.	FR06
F-PC1	Add component link	Authorized product-management role	Define a parent-component pair and required quantity.	FR09
F-PC2	Validate cycle prevention	System	Reject direct or indirect BOM cycles.	FR10, BR11
F-PC3	Calculate multi-level requirement	System	Calculate total component requirements along valid product-structure paths.	Section 6
F-SM1	Log transaction	System	Create exactly one StockMovement for each successful opening, receipt, issue or adjustment.	FR08, BR12
F-SM2	Reject invalid issue	System	Reject an issue that would make Inventory.Quantity negative.	BR12
F-RA1	Create alert	System	Open an alert when Quantity < ReorderLevel and no alert is already open.	FR11, BR14
F-RA2	Close alert	System	Set ResolvedAt when Quantity recovers to ReorderLevel or above.	FR11, BR15
F-RA3	Prevent duplicate alerts	System	Keep the existing open alert while the low-stock condition persists.	BR14

2.7 User Permission Definitions

Actor / Role	Allowed Functions	Permission Scope
Purchasing Staff	F-WH1, F-PR1, F-SU1, F-SU2, F-SP1, F-SP2, F-SO1, F-SO2, F-SO3, F-SO4	Supplier, product, pricing and purchasing operations.
Warehouse Staff	F-WH2, F-WH3, F-SI2, inventory issue/adjustment operations	Warehouse receipt and inventory-side operations.
Logistics Staff	F-SH1, F-SH2, F-SH3, F-SH4, F-TE1	Shipment creation, allocation, dispatch, arrival and tracking.
Inventory Manager	F-WH2, F-PR3, F-SO4; review RestockAlert	Inventory monitoring and replenishment decisions.
System / automatic processing	F-WH4, F-IN1, F-IN2, F-IN3, F-OI2, F-SI1, F-PC2, F-PC3, F-SM1, F-SM2, F-RA1, F-RA2, F-RA3	Automatic validation, balance update, BOM checks and alert logic.

III. Functional Requirements

ID	Function	Requirement
FR01	Warehouse management	Store the code, name, address, city, country, and capacity of every warehouse.
FR02	Product management	Store product details, base unit, and product type.
FR03	Supplier management	Store contact details and the products currently offered by each supplier.
FR04	Supplier orders	Create an order for one supplier, one destination warehouse, and several product lines.
FR05	Partial deliveries	Allocate one order item to several shipments without exceeding the ordered quantity.
FR06	Shipment tracking	Store a tracking number, dispatch date, expected date, arrival date, status, and chronological events.
FR07	Inventory management	Store the current quantity and reorder level for each warehouse and product pair.
FR08	Stock movements	Create one StockMovement whenever a balance is opened, received, issued, or adjusted.
FR09	Product structure	Store the parent product, component product, and required quantity at every level.
FR10	BOM cycle prevention	Prevent a product from containing itself directly or indirectly.
FR11	Restock alerts	Create one open alert when Quantity is below ReorderLevel and close it after stock recovers.

IV. Business Rules

4.1

ID	Business Constraint
BR01	Each Inventory record belongs to exactly one Warehouse and one Product. A warehouse or product may have zero or many inventory records. WarehouseID and ProductID must be unique as a pair.

ID	Business Constraint
BR02	Supplier and Product have a many-to-many relationship through SupplierProduct. Each supplier product record stores the current UnitPrice and LeadTimeDays.
BR03	Each SupplierOrder refers to exactly one Supplier and one destination Warehouse. A supplier or warehouse may have zero or many orders.
BR04	A draft order may have no items, but a submitted order must have at least one. Each SupplierOrderItem belongs to exactly one order and one product, and a product cannot repeat in the same order.
BR05	The order supplier must have a SupplierProduct record for every ordered product. UnitPrice on SupplierOrderItem is the agreed historical price and does not change with the current supplier price.
BR06	An order may have zero or many Shipments. Each Shipment belongs to exactly one order and inherits the destination warehouse from that order.
BR07	A draft shipment may have no items, but a dispatched shipment must have at least one ShipmentItem. Each ShipmentItem refers to a SupplierOrderItem from the same order as the shipment.
BR08	A SupplierOrderItem may be divided among several shipments. The total quantity of non-cancelled ShipmentItems must not exceed the ordered quantity.
BR09	Each TrackingEvent belongs to exactly one Shipment. A shipment may have zero or many events. TrackingEventID distinguishes events that occur at the same time.
BR10	Product contains Product is a recursive many-to-many relationship. ProductComponent stores the parent product, component product, and a QuantityRequired value greater than zero.
BR11	A product must not contain itself directly or indirectly. A product without components is a leaf item, while a component may still contain lower-level components.
BR12	Inventory.Quantity and ReorderLevel cannot be negative. A stock issue is rejected if it would make Quantity negative. Each successful change creates exactly one StockMovement.
BR13	A Receipt movement refers to exactly one ShipmentItem, and each ShipmentItem may be received at most once. Other movement types do not refer to a shipment item.
BR14	Low stock occurs when Quantity is less than ReorderLevel. Each Inventory record may have at most one unresolved RestockAlert. Repeated low-stock updates must not create duplicate alerts.
BR5	When Quantity is greater than or equal to ReorderLevel, the open alert is resolved. If Quantity later falls below the threshold, a new alert period begins. A change to ReorderLevel must also be evaluated.

4.2 Relationship Definitions, Cardinality and Participation

ID	Relationship	Cardinality	Participation / Meaning
R01	Warehouse–Inventory	1 : 0..N	Every Inventory belongs to exactly one Warehouse.
R02	Product–Inventory	1 : 0..N	Every Inventory belongs to exactly one Product.
R03	Supplier–SupplierProduct	1 : 0..N	Every SupplierProduct has one Supplier.
R04	Product–SupplierProduct	1 : 0..N	Every SupplierProduct has one Product.
R05	Supplier–SupplierOrder	1 : 0..N	Every order has exactly one Supplier.
R06	Warehouse–SupplierOrder	1 : 0..N	Every order has exactly one destination

			Warehouse.
R07	SupplierOrder–SupplierOrderItem	1 : 0..N	Draft may be empty; submitted order has at least one item.
R08	Product–SupplierOrderItem	1 : 0..N	Every order item selects one Product.
R09	SupplierOrder–Shipment	1 : 0..N	Every Shipment belongs to one order.
R10	Shipment–ShipmentItem	1 : 0..N	Draft may be empty; dispatched shipment has at least one item.
R11	SupplierOrderItem–ShipmentItem	1 : 0..N	Every shipment item is allocated from one order item.
R12	Shipment–TrackingEvent	1 : 0..N	Every tracking event belongs to one Shipment.
R13	Parent Product–ProductComponent	1 : 0..N	Every component link has one parent Product.
R14	Component Product–ProductComponent	1 : 0..N	Every component link has one component Product.
R15	Inventory–StockMovement	1 : 0..N	Every movement belongs to one inventory balance.
R16	Inventory–RestockAlert	1 : 0..N	Every alert belongs to one inventory balance.
R17	ShipmentItem–StockMovement	0..1 : 0..1	Only Receipt uses this link; a ShipmentItem is received at most once.

4.3 Key Business Constraints

Constraint	Definition	Rule
Inventory uniqueness	One Warehouse–Product pair has at most one Inventory record.	BR01
Supplier-product validity	The order supplier must offer every ordered product.	BR05
Order uniqueness	A Product cannot repeat in the same SupplierOrder.	BR04
Partial shipment limit	Non-cancelled shipment allocations cannot exceed ordered quantity.	BR08
Shipment-order consistency	ShipmentItem and Shipment must belong to the same SupplierOrder.	BR07
Inventory non-negativity	Quantity and ReorderLevel cannot be negative; invalid issues are rejected.	BR12
Receipt uniqueness	A ShipmentItem may be received at most once.	BR13
Recursive structure validity	No direct/indirect Product cycle; QuantityRequired > 0.	BR10–BR11
Restock alert uniqueness	At most one unresolved RestockAlert per Inventory.	BR14
Alert lifecycle	Low stock opens an alert; recovery closes it; a later low-stock period opens a new one.	BR14–BR15

4.4 Important Modelling Clarifications

Entity vs Actor: Warehouse, Product, Inventory, Supplier and the other listed objects are database entities. Purchasing Staff, Warehouse Staff, Logistics Staff and Inventory Manager are actors/roles.

Current state vs history: Inventory stores the current operational balance, while StockMovement stores the history of successful inventory changes and RestockAlert stores alert periods.

Current price vs historical price: SupplierProduct.UnitPrice is the current supplier quotation; SupplierOrderItem.UnitPrice is the agreed price captured for that order.

User action vs system action: A user initiates a valid business action; the system then performs required automatic updates and validations.

Recursive roles: ParentProductID and ComponentProductID refer to the same Product entity but represent different roles.

Phase 1 vs Phase 2: Phase 1 defines the conceptual model and business constraints. DBMS-specific SQL constraints, triggers, indexes and implementation details are deferred to Phase 2.

V. Extended Entity Relationship Model

The diagram below is the complete Phase 1 EER model. The symbol 1 means exactly one record, and 0..N means zero to many records. Relationships R01 through R17 are defined precisely in Section 4.

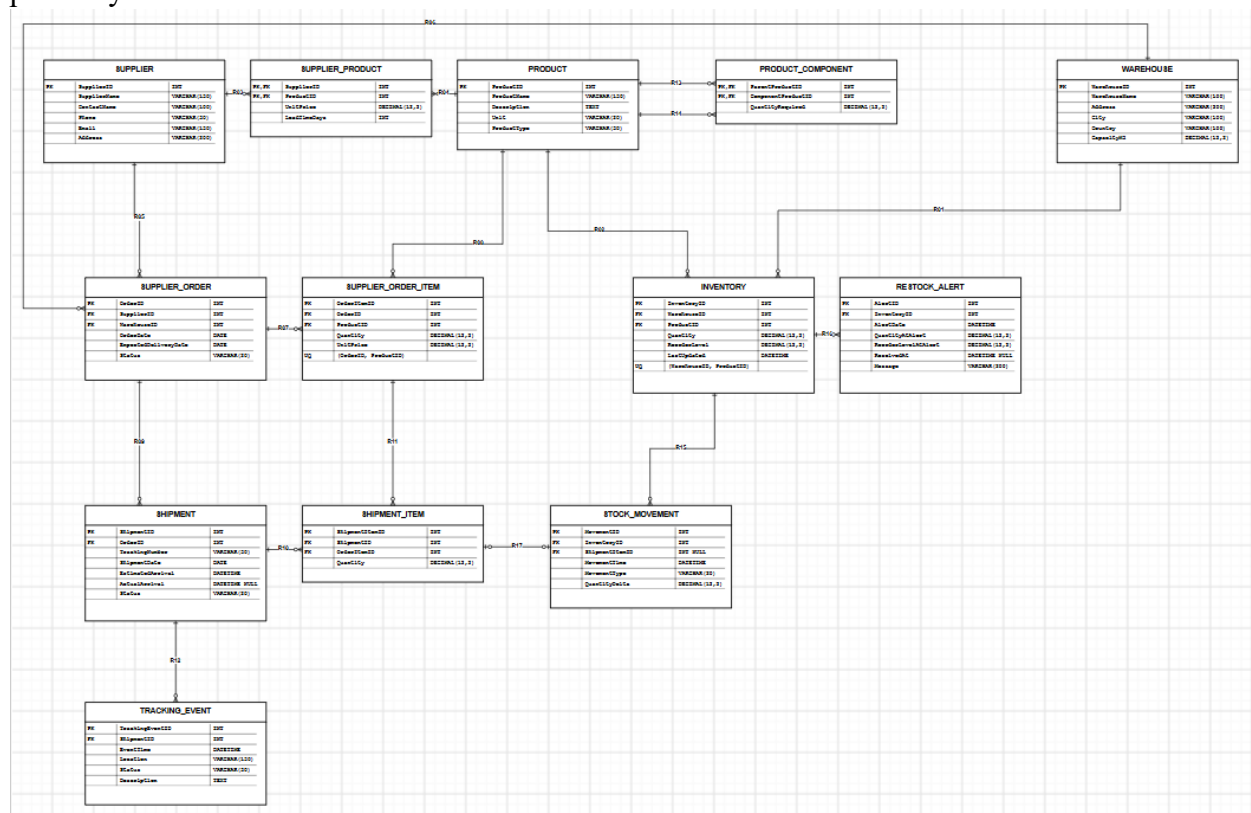


Figure 1 Complete EER model for warehouse logistics product structures and restock alerts3

